

A Multi-Site Label Trap in Shape-Based Endometriosis Detection

What female pelvic-organ shape does (and does not) encode



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424

women across two public MRI datasets

78% vs 15%

endometrioma annotation prevalence, site D1 vs D2

0.826

AUC within site D2 (95% CI 0.67–0.94)

0.274

transfer AUC D1→D2 (CI entirely below 0.5)

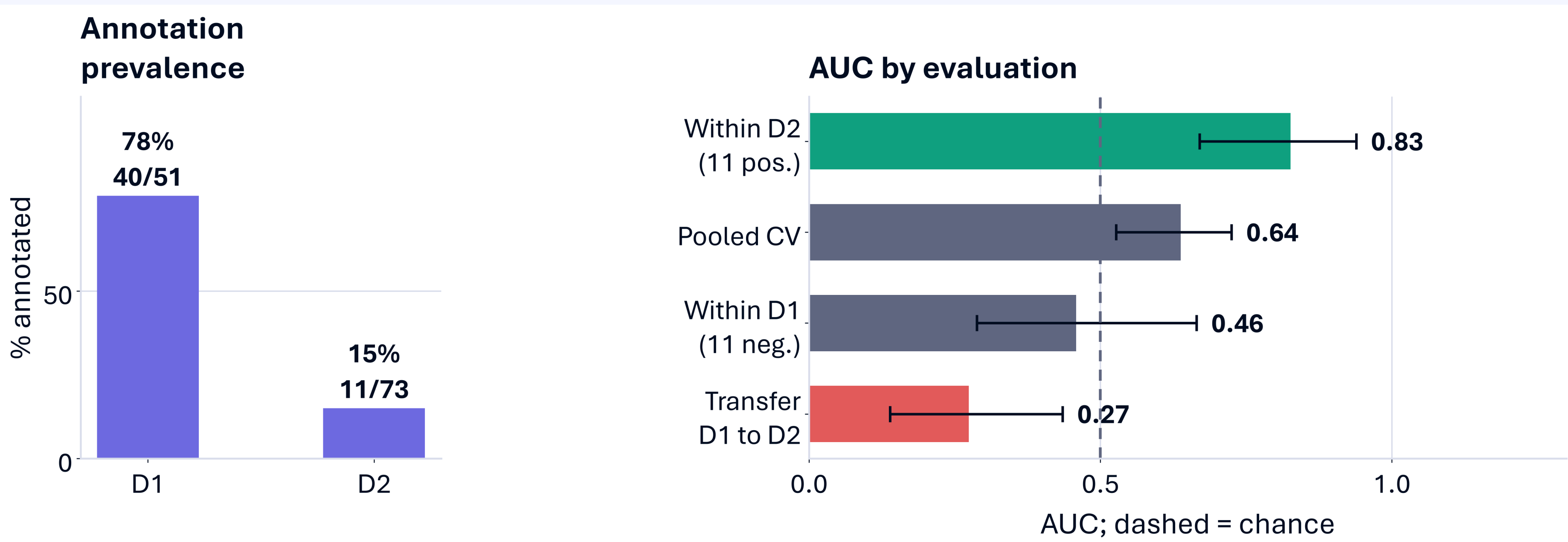
Three questions

- Does uterine shape encode age, and is it the uterus or the fibroids talking?
- Can shape predict which patients carry an endometrioma annotation, and does that survive a site change?
- How stable are shape descriptors when a different rater segments the organ?

Data & methods

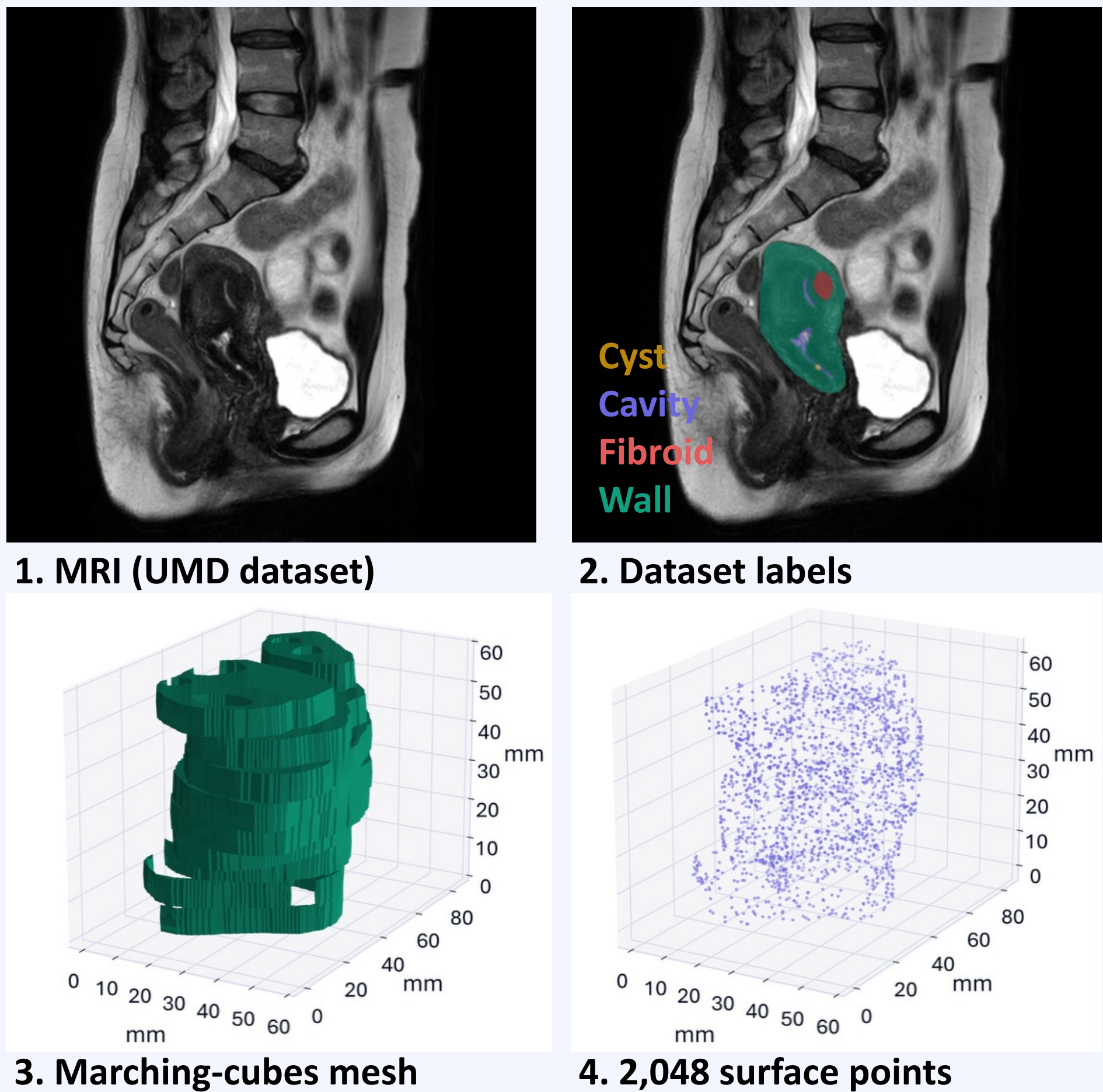
- UMD: 300 women, ages 21–86.
- UT-EndoMRI: 124 patients, two sites, three raters at D1.
- XGBoost (gradient-boosted trees), subject-level 5-fold CV, bootstrap 95% CIs.
- Size descriptors grow with the organ (volume, area, overall dimensions); scale-free shape captures its form regardless of size (sphericity, elongation, aspect ratios).

The label trap: one model, four verdicts



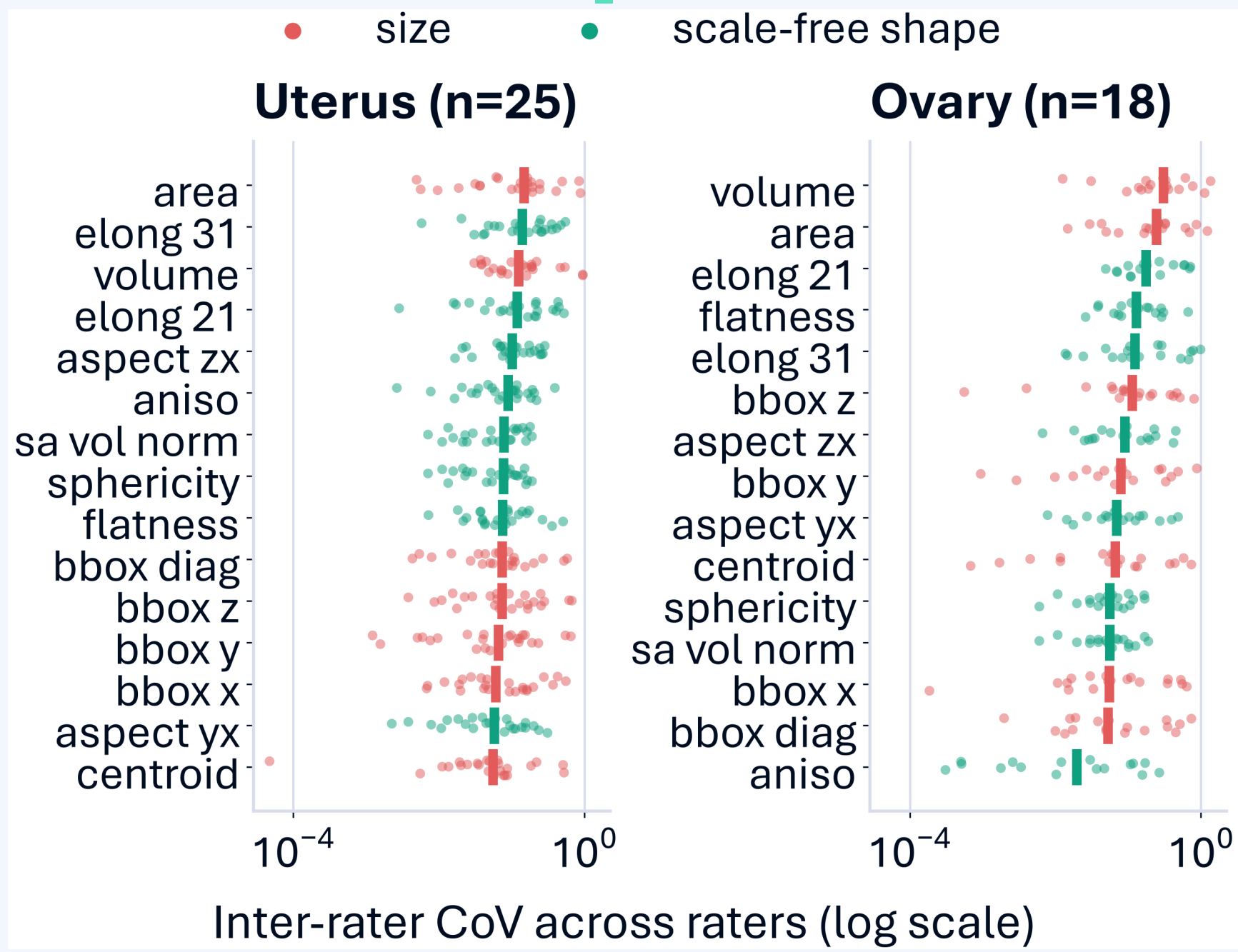
- The endometrioma label marks the presence of an annotation file. D1 annotated endometriomas in 78% of its patients, while D2 annotated only 15%. These are annotation habits, not related to true prevalence.
- Within D2 the annotation is predictable from shape: AUC 0.826 (95% CI 0.67–0.94, permutation $p = 0.004$).
- Pooled cross-validation conflates site with pathology (0.638); transfer D1→D2 collapses to 0.274 with the CI entirely below 0.5.
- Any multi-site study where annotation completeness tracks the target risks this confound. Report per-site prevalence and cross-site transfer.

The pipeline

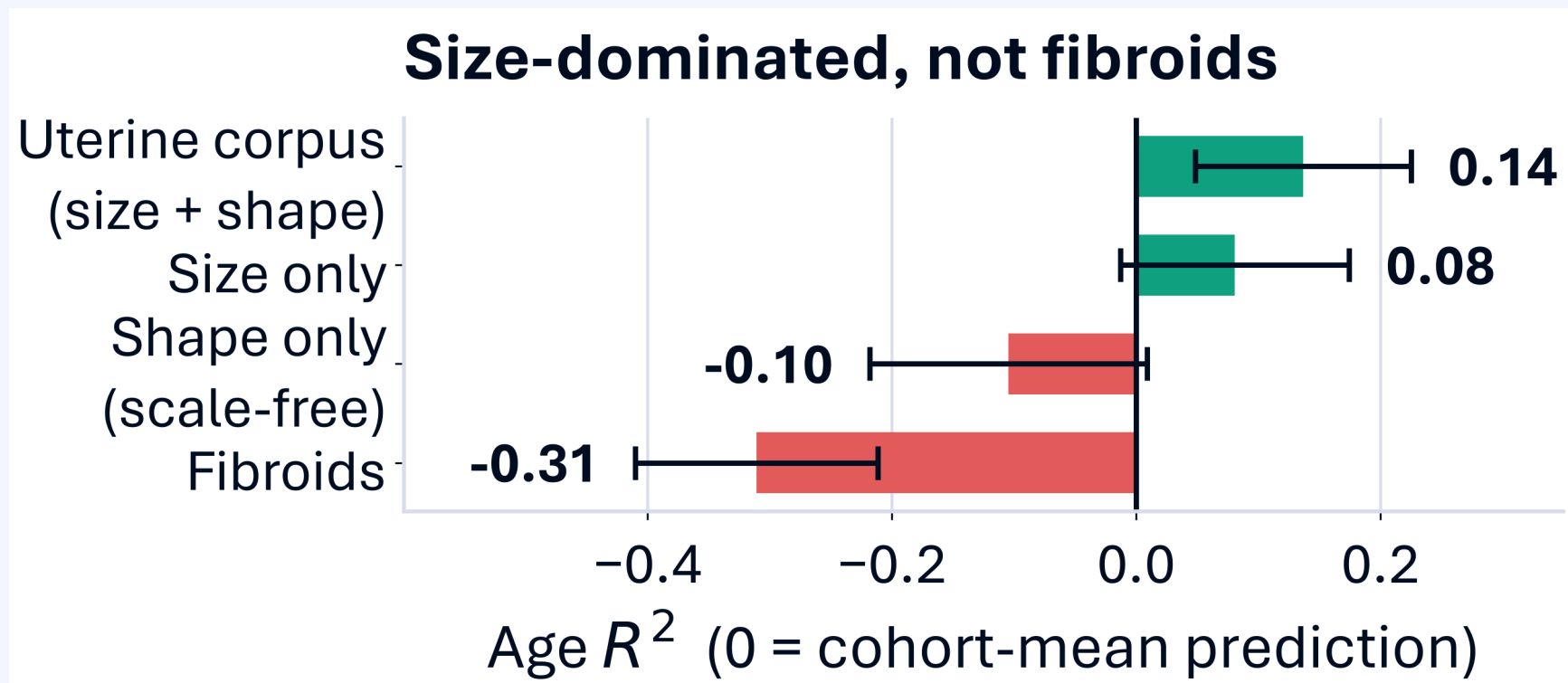


Rater stability

Age: size-driven

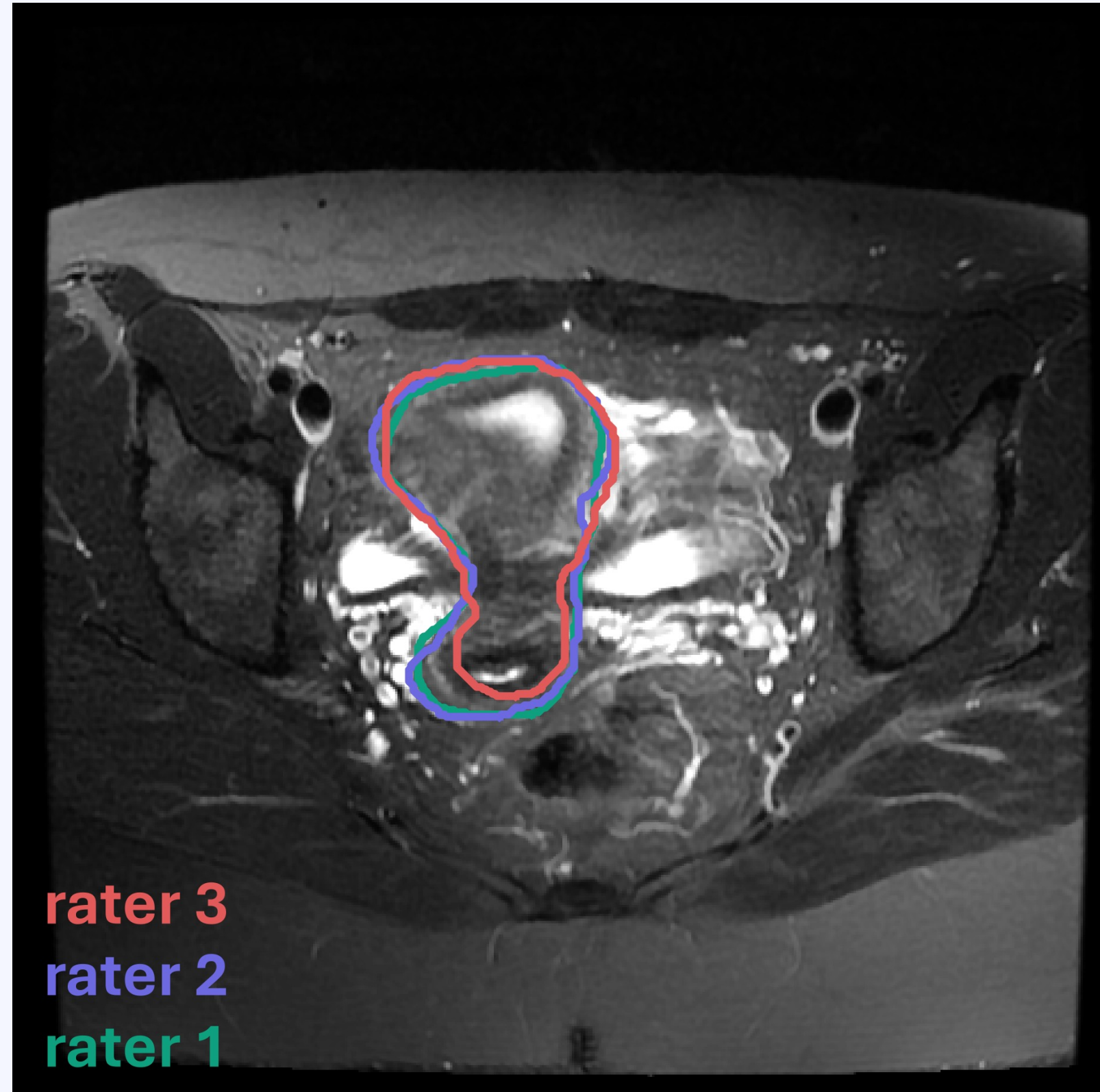


Averaged over all descriptors, shape is more rater-stable than size; the two groups overlap, and volume and area vary most across raters.



MAE 8.9 yr vs 10.2 baseline; scale-free shape alone falls below the mean; fibroids carry nothing.

Three raters, one uterus



Three raters outline the same uterus (site D1); the boundary is where they disagree.

Limitations

Modest cohorts; annotation labels; 11 positives behind the headline AUC; raters from one site (25 uterus / 18 ovary subjects).

Before you pool

- Check label prevalence per site.
- Test cross-site transfer.
- Report a size control next to any shape biomarker.



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